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The Unconscious as Infinite Sets

An Essay in Bi-logic

With a New Foreword by
Eric Rayner

ROUTLEDGE

THE UNCONSCIOUS AS INFINITE SETS



An Archangel
Byzantine Mosaic from Khora Museum, Istambul

The three pairs of wings together with the voluminous body give the impression that the latter actually corresponds to the bodies of three birds; hence its density and abundance. In fact, it looks as though the three birds had been catapulted into one portion of space, which would contain them, leaving out the wings.

Compare this mosaic with the four-dimensional tetrahedron of Fig. 3 (page 412) which, if represented in three-dimensional space, 'yields' several three-dimensional tetrahedrons.

Alternatively, the picture conveys the impression of the interpenetrability of volumes, in conformity with the principle of symmetry. Both interpretations suggest that the artist intuitively conceived the archangel as a being submitted to laws which are beyond those of Aristotelian logic or of three-dimensional space but which could be 'translated' or 'unfolded' into these laws.

On the other hand, the head of the archangel, almost hidden by the wings, suggests a space of dimensions higher than three which is 'wrapped' in three-dimensional spaces: the wings. The well-delimited features of the face and the penetrating eyes, in contrast, suggest a bi- or three-dimensional sharpness which is characteristic of consciousness. (Photograph by Sergio Larrain)

THE UNCONSCIOUS AS INFINITE SETS

An Essay in Bi-Logic

Ignacio Matte Blanco

with a new Foreword by

Eric Rayner

Routledge
Taylor & Francis Group
LONDON AND NEW YORK

Originally published in 1975 by
Gerald Duckworth & Company Limited

Fist published 1998 by H. Karnac (Books) Ltd.

Published 2018 by Routledge
2 Park Square, Milton Park, Abingdon, Oxon OX14 4RN
711 Third Avenue, New York, NY 10017, USA

Routledge is an imprint of the Taylor & Francis Group, an informa business

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British Library Cataloguing in Publication Data

A C.I.P. record for this book is available from the British Library.

ISBN 9781855752023 (pbk)

In memory of my father

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Foreword

This book is undoubtedly Matte Blanco's most fundamental work. It has greatness and has been in print for nearly a quarter of a century, but frankly it is still shamefully ignored by most psychoanalysts and therapists. It is huge, and many lose courage when Matte says that it is written for psychoanalysts as well as for mathematical philosophers. But he was, through and through, a psychoanalyst with a great deal of clinical experience, and his friendly enthusiasm will carry you with him through many meeting places of abstract ideas. Once his basic approach is grasped, the book is 'unputdownable'. With his rare facility in psychiatry, psychoanalysis and mathematical logic, he produced, perhaps with the exception of Bion, the only fundamental account so far of the logical structures involved in emotions and the unconscious.

How can one get into his way of thinking? Perhaps a glance at his characteristic way of doing things can help. Matte was born and educated in Chile; he trained as a psychoanalyst and psychiatrist in England, worked in the United States, returned to Chile as a professor of psychiatry and psychoanalyst, then moved over thirty years ago to Rome to teach and practise. He was a high-born Spanish-Chilean aristocrat, with a proud lineage going back to heroes—and, no doubt, villains—hundreds of years ago. At the same time, he could get on enthusiastically without arrogance with anyone. He once said to me, "You know, Eric, I think people can best understand what I am saying by recognising that I am a Latin and thus very emotional. But, when I was about five, my father gave me a nickname which in English would be something like "little hair-splitter". That is me—very emotional and logical at the same time.' This is what his bi-logic is about: it asks, what logic is in emotions, and what place does emotion have in logical thought?

The starting point of bi-logical thinking is described by Matte in the following pages. Put roughly, I think it sees the mind in a binary way, performing two utterly fundamental functions. First, when awake we must continuously register the *'sameness'* of things around us in relation to things remembered from the past, such as 'This here is the same as that thing I saw the other day' etc. In effect, this use of *memory*, one of the great feats of humankind, saves

an impossible labour of everything met being a new problem. We, like all mammals, are 'similarity-finding' animals. Matte highlights this facility by introducing what he calls 'the principle of *symmetry*' (of sameness or similarity, in other words).

Next, in order actually to find one's way around the world, we must continuously estimate *difference relations* between the thing around us, such as 'there are traffic lights *ahead* and a car is *behind* me'. 'Ahead' and 'behind' here indicate specific difference relations. Matte emphasises the omnipresence of these relations by introducing the 'principle of *asymmetry*'. These two functions—the experience of symmetries and asymmetries, of similarities and differences—continuously interact, and the study of their interplay he called "bi-logic".

Now, moving to psychoanalysis: when fully *conscious* and discriminating intellectually, we function with an integrated and consistent *interplay* of discrimination of asymmetries together with registering similarities or symmetries. However, when emotional, we grasp similarities very readily, although often at the expense of careful discrimination; prejudices may occur, but quick reactions—which are vital in social interchange and in emergencies—do become possible.

From emotionality, Matte the psychoanalyst turns particularly to the unrepressed unconscious. Here, he says that things that are discriminated as in a *difference relation* at a *conscious* level tend to be registered as *the same* at *unconscious* levels. This he calls 'the principle of symmetry'. Put in another way, he says that 'symmetrization'—or perception of similarities—'rules' in the unconscious.

From this he goes from one logical step to another, thinking about unconscious and emotional processes. One *tour de force* is to draw together Freud's five rather disparate characteristics of the unconscious in the light of the interplay of symmetry and asymmetry; there are many other insights. The basic core comes in his first 150 pages; one tends to feel a little mad there, getting into his unusual way of thinking, but, once this is grasped, the rest of the book is a fascinating exploration. When able to think in Matte's way, I have found that actual, everyday clinical work becomes that bit easier and more satisfying. To get the most out of it, one needs to enjoy being both intensely intellectual and highly emotional at the same time.

The book is now about a quarter of a century old and still not well understood; bi-logic is nowhere required reading for students. But within a small yet world-wide group of us, who took to Matte's bi-logic, quite a lot has happened since 1975. Here are some recollections.

Matte himself died aged 86 in 1995 and had been unable to work for some years before that. His last work in English was *Think-*

ing, Feeling and Being (1988). This is not an easy read. He, an old man in a hurry, is meditating fast and comes tumbling out with dozens of fascinating ideas—but he darts quickly on. He begins by addressing the *structure* of the unconscious. However, he is *not* referring to Freud's structural point of view—of id, ego and super-ego. He is addressing the differences between the logical structures of conscious and unconscious thought. It is really about Freud's topographic point of view. In contrast to the present book, in *Thinking, Feeling and Being* he is determined to concentrate upon clinical matters. But Matte still cannot resist highly abstract ideas. For instance, he meditates upon the antinomy of 'being and happening'. This will be too metaphysical for some; even so, he is always refreshingly adventurous—as long as one can keep up.

Next, he is happily clinical, plunging into a careful study of the logical structures of projection and introjection and of the internal world. He remembers his seminars as a student with Melanie Klein in the 1930s. Admiring, but not uncritically, he recalls what a profound intuitive sense of the unconscious Klein possessed; but she was tied to conscious-level concepts that were inadequate to communicate about the dark regions she was in. Incidentally, the two most well-known investigators of logic and the unconscious—Wilfred Bion and Matte—had links with Klein. Bion's daughter, Parthenope Bion-Talamo, has written a thesis comparing the ideas of the two men.

Matte then moves naturally into the logical structure of projective identification. From this he goes on to propose a new way of looking at the structure of the mind. This suggests that the *unrepressed* unconscious, as distinct from that which has been repressed, may be seen in terms of a *stratified bi-logic structure.* Here, conscious levels use an organised interplay of asymmetrical and symmetrical thought; then, with emotionality, looser symmetrizations come into play, while deeper unconscious levels manifest much higher proportions of symmetry. Later in the book he uses the biological work of his long-term friend John Bowlby, the originator of attachment theory, to point out the *survival importance* of symmetrizations at emotional and unconscious levels.

Matte goes on to investigate some forms of disturbance that have hardly been thought of before: states of intellectual *frenzy* that manifest different forms of upheaval of spatial and temporal thought structures. Some might say that these were obsessional activities, but this would miss the quality of frenzy of *logical turbulence* that is being addressed. There are many vivid and fascinating clinical examples. Now well into his stride, logical and clinical thinking come together about the notion of an object, space, dimension, outside, inside. Lastly, Matte engages with the notion of internal world, past, present and future.

This book, like his first, is enthralling, tumbling with new thoughts. It was published over a decade ago but has not, however, caught on. Perhaps it is too logical for the psychoanalyst and too clinical for the logician. Perhaps Matte hops about and does not bother enough where the reader is. But its ideas deserve more attention than they have received: one day, some of us must get together and start a new creativity out of it.

I will now turn to look briefly at what the rest of us have made of his work in the past decades. Matte spent his last thirty years in Rome, and his family still live there; his widow, Luciana Bon de Matte, is a well-known training analyst in Rome and an active member of our group. It is in Italy that Matte has been most known and used. Alessandra Ginzburg writes the following from there.

> Matte Blanco rouses almost more interest among humanist intellectuals than psychoanalysts. Philosophers such as Remo Bodei and Giorgio Gargani, or scholars of literature such as Francesco Orlando and Guido Paduana, put his thoughts to use in their studies. Among his more direct psychoanalytic pupils several teach psychiatry at University, one holds a Chair of philosophy, while others are exclusively practising psychoanalysts. Prof. Pietro Bria is translator and editor of Matte's work, but above all is the animator of every cultural initiative. The cultural fervour, which Matte Blanco's approach provoked, lasted after his death: three important congresses based on his work have taken place. The first in 1981 saw a vast delegation of elements of Italian culture. The second in 1995 was a memorial day where psychoanalysts presented their clinical reflections, while Francesco Orlando gave examples of the literary application of symmetrical logic. The third congress in Amalfi in 1997 was on the subject of 'The Fundamental Antinomy of Being', and philosophers, psychoanalysts and other scholars participated together. A number of studies have been and are to be published. The need for international links is recognised and being sought by several of us.

From Chile, Juan Francisco Jordan says that before the mid-1980s the work of Matte was utterly unknown to Chilean psychoanalysts and psychiatrists. Then a small study group was formed and has continued since. Two symposia have been held, and the papers given at the first one in 1991 have been published. Matte's work is starting to spread among psychiatrists who find his bi-logic extremely useful in understanding schizophrenic psychosis. Nowadays, he is acknowledged by many Chilean psychoanalysts as one of the major contributors to psychoanalytic thought world-wide.

In Canada, Claudio Duran, a philosopher, has formed a small bi-logic study group, mostly of philosophers. Coming originally from Chile, he has over many years carried out a study of the propaganda methods used by the right-wing press there at the time of the 1973 coup when Pinochet came to power. Duran explicitly

and centrally used bi-logic psychoanalytic theory to explain the mass impact of the persuasion involved. His findings have been published and are well known in Chile today.

It was also Claudio Duran who was instrumental in the starting of a bi-logic discussion group in England. I myself had published a paper about bi-logic in 1981 and, soon after, met Ignacio Matte Blanco; we quickly became friends and remained so till he died. About fifteen years ago he told Duran to contact me when he was next passing through London. He did, and he told me that it was about time a regular bi-logic discussion group was started. London was an obvious centre for other countries. We have met regularly two or three times a year since then. We are mostly Londoners, but nearly always someone flies in. The numbers have been kept down to about twenty people so that we can meet in each others' homes; there has never been shortage of volunteers to read papers—several dozen have been given by now.

The group has intentionally been interdisciplinary: it includes one or two philosophers, three or four psychoanalyst, several psychiatrists, child psychotherapists, a group analyst, a lawyer, a linguistician, a mathematician, a publisher, and a priest. Though most live in London, people from the north of England, Ireland, Italy, Germany, Sweden, Canada, and the United States also belong and attend quite often. The meetings are quiet but often unforgettable; civilised human beings are together at their best. Apart from bi-logic, the essential ingredient is open-minded enjoyment of abstract thinking. Most psychoanalysts and therapists show no interest in Matte; it is partly by default that the London group is so interdisciplinary. The same has happened in Italy.

Here is a brief list of what has been discussed in the London group. My friend in it from the beginning has been the psychoanalyst Gerald Wooster; it was he who brought most of us together. He has also fascinated us on many occasions with the bi-logical devices used by William Shakespeare in his plays, and by their appearance in opera and in the Bible. Wooster, Rodney Bomford, who is an Anglican priest and ex-mathematician, and Judith Elkan, an amateur Talmudic scholar and child therapist, have gripped us with bi-logical discussions of the book of Genesis. My other early psychoanalytic colleague, Margaret Arden, started us off clinically and sociologically by comparing Matte and Bateson; more recently she has been interested in religious experience and psychoanalysis. Alejandro and Paulina Reyes, child psychiatrist and child therapist, and also Michael Gopfert, a psychiatrist and therapist, are interested in the experience of *value*—financial versus personal—using bi-logical methods. Jim Wilson, a psychotherapist, speaks bi-logically about experiences in groups.

Joan Maw, a linguistician, has naturally been interested in verbal symbolisation and symmetrization. She has shown how the words

of different languages emphasise different modes of interpersonal relationships, which can usefully be highlighted by a bi-logical ear. More recently she has turned her bi-logic to visual symbolism. Rodney Bomford has been our most prolific member. He thinks that bi-logic opens ways between psychoanalytic thinking, literature and theology. He has shown how Freud's five characteristics of the unconscious have similarities with the characteristics of the Christian God. His bi-logic has also elucidated many forms of rhetorical speech and shown how they express such experiences as infinity, eternity, symbolism and paradox and also the reality and fantasy in stories. Ian Mordant, a mathematician also concerned with psychotherapy, has not just read papers to us—many of us would never have understood Matte's mathematical logic without his help. Ian's first major formal contribution was to show how 'extraordinarily *abstract* the unconscious is'. He sees primary process as the outcome of the mental activities in abstraction. These are essential for the quick reactivity necessary for survival. More recently, he and I have been interested in the logical structures in the personal experience of fairness and justice. I have been concerned to show how these are implicit in much therapeutic activity; he has concentrated upon the functions of splitting and symmetrization in unfairness and injustice.

Tony Sporrong, a psychiatrist and therapist in Sweden, has more recently qualified as a philosopher. He considers that the special *object of knowledge* for psychoanalysis is the *unrepressed* unconscious; when this is clear, psychoanalysis can be better differentiated from general psychology, whereas up to now they have competed and muddled each other. Ross Skelton, a university philosopher and logician from Ireland, is also a therapist there; he first contributed by engaging publicly with Matte to show that his theory was contradictory as a metapsychology. A lively debate ensued which, Skelton thinks, eventually showed that bi-logic was primarily intended as a clinical theory and not a general metapsychology. He thinks that Matte's main achievement has been to take Cantor, the discoverer of the mathematical theory of infinity, seriously. In adopting Cantor's theory of the infinite, Matte has helped to establish a '*poetic science*'. Skelton's aim more recently has been to modernise the logic in bi-logic and also to link Matte with the work of both Bion and Lacan.

Klaus Fink, a psychoanalyst living now in Germany, has been our main exponent of the clinical uses of bi-logic; his papers are widely published. Carmen Fink, Klaus' wife, a lawyer, has written on bi-logic and intentionality in criminal offences, and also on fairy stories. Richard Carvalho, a Jungian analyst, has, like Bomford, investigated the logical structures in ordinary thoughts. Jim Rose, a psychoanalyst, has worked on distortions of time; while Alan Shuttleworth, together with Judith Elkan and Paulina Reyes, all

child therapists, have elucidated *childhood autism* bi-logically. I myself, a psychoanalyst, have put most of my bi-logical efforts into elementary expositions to others. I have thus written a basic introductory book on the subject (1995b). I have also been interested, amateurishly, in bi-logic and chaos theory. In a more scholarly way, I have been concerned with the experiences of fairness and justice and their importance in the therapeutic process (1995a). I, together with Ian Mordant and other bi-logicians, am likely to be immersed in this subject for the rest of my days.

Bi-logical discussion came to the United States more recently. The first to become interested was Itamar Yahalom, who started by reading the original edition of *The Unconscious as Infinite Sets*; he had worked with Bettelheim for many years and is a psychoanalyst in Los Angeles. There is now an active bi-logic group of psychoanalysts and therapists working and teaching there, in particular Laurie Ryavec, Carole Morgan and Bill Edkins. While the British group is primarily interdisciplinary, the Californians are more actively centred upon clinical theory. Though their group is much younger than ours in London, they are likely soon to be leading the field in clinical matters as well as in numbers. You will notice that most of us have applied bi-logical theory to other areas rather than engage with developing the logical origins. We must go on with this.

At the time of writing we have had very few quarrels and are starting an 'international bi-logic group'. What do we have in common that makes for this blessed peacefulness? I think that it is, first, an enjoyment of abstract ideas. But just as central is being unashamedly emotional. So an intense 'intellectual emotionality' brings and keeps us together. The emotional communality allows us to be informally democratic. This, together with the enjoyment of abstraction, keeps us intellectually open-minded. It is a rare and blessed combination which, I think, all of us value quietly but deeply. I, for one, am proud to have known Matte and to be one of our group.

Eric Rayner

February 1998

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Preface

This book is written for psycho-analysts as well as for mathematical philosophers. I realise that the requirements of these two groups of readers are divergent and that to a certain extent – if not inevitably – they appear mutually exclusive. Unfortunately, however, there is nothing I can do to avoid this difficulty, for it seems to me that it is the subject itself that must be considered in terms of both approaches. On the one hand, I am firmly convinced that a reformulation of psycho-analysis in terms of logico-mathematical concepts will profoundly revolutionise philosophical thinking and greatly change some of the basic tenets upon which Western philosophy and science rest, and this book is an attempt, however imperfect, at such a reformulation. In it I propose guidelines which I consider essential for this purpose. As will be seen, some of the notions employed in my presentation – such for instance, as those of relation and correspondence or mapping – are at the very foundations of both logic and scientific knowledge. The use in psycho-analysis of precise mathematical concepts permits, I believe, the development of a new and wider view of the mind, of a greater degree of precision, and leads to a synthesis and a unity of apparently widely disparate subjects.

On the other hand, I am confident that if the psycho-analytical reader is willing to spend the time required to become familiar with the arguments developed – and, I may add, the book is self-contained, in the sense that it presents all the logico-mathematical notions required to understand it – he will find that these arguments, in their turn, will increase the subtlety of his understanding of patients and in various other ways considerably enrich his clinical work. For, however remote from clinical practice these ideas may appear, they have all emerged from contact with the patient.

This book, if considered as the result of many years of psycho-analytical observation, of sustained reflection on its meaning and of an effort to draw the consequences that follow from it and from various other discoveries of psycho-analysis, may be regarded as the culmination of a point of view. But it is also, in my opinion, a starting-point for new developments, of which three now preoccupy me. The first refers to the question of God. If the findings of Freud on this subject are reformulated in terms of the approach presented

here, then the immanent notions about God, interpreted as bi-logical experiences, lead to interesting perspectives. The second concerns the so-called death instinct which, if viewed again as a bi-logical conception, instead of as an instinct, seems to provide fresh insights into the nature of life.

Finally, if the question of knowledge is considered in terms of the two modes of being in man, certain new epistemological principles become immediately apparent. In particular, every time we know a concrete object we also know in some way the complete class or set to which it belongs, and we know it or live it as homogeneous and indivisible: individual and class are the same thing. This, in turn, leads to a way – a very old way – of looking at being. I am referring to the connection – which I have only recently discovered[1] – that the formulation proposed in this book has with the views of the Eleatic philosophers. The resemblances are striking; and at the same time it is obvious that the paths followed are radically different. Parmenides started from purely logico-metaphysical considerations, whereas this book begins from the observation of psychological reality with the help of psycho-analytic tools and conceptions, reformulated in logico-mathematical terms. On the other hand, if 'the doctrines of Parmenides offend against common sense by denying the reality of change and diversity', as Hussey (loc.cit., p. 99) quite rightly points out, this seems to be due to his considering exclusively only one aspect of the question; whereas careful reflection actually shows that the facts discovered by psycho-analysis can be properly understood and formulated in terms of *two* co-existing modes of being in the world: a homogeneous indivisible mode and a heterogeneous dividing one. (For details see Chapter 28, where the bases for an epistemological approach are suggested, but not developed; and in particular the comments on Wittgenstein.)

Viewed in this light, it seems that some of the ideas of Parmenides and Zeno could be incorporated into the very centre of the philosophical and scientific conception of man and of the nature of knowledge, instead of being left on the outside – as they seem to have been for twenty-five centuries – as fascinating but deeply disturbing manifestations of the Greek genius. It is strange that, of all approaches, it should have been psycho-analysis that has contributed to a re-assessment of Parmenides' and Zeno's apparently fantastic and apparently highly metaphysical conceptions. Yet psycho-analytic observation, in the formulation proposed here, can rescue these ideas from the remote region to which they have been relegated, and show that they play an important role in our ordinary conception of the world, provided that they are integrated, as parts, into a wider conception of man.

[1] Thanks to Hussey's lucid book (*The Presocratics*, 1972). See also Calogero (1967) and Cornford (1939).

So far as the presentation of the bi-logical point of view in psycho-analysis is concerned, I feel that the argument is sufficiently developed here. Anyone who is interested only in this subject may read no further than Part VIII. However, gradual reflection about bi-logic has led me in new directions when considering the mind in terms of multidimensional space, as I have suggested and discussed in various connections throughout the book. It seems too early to decide what the ultimate relation between bi-logic and the view of the mind presented here in terms of multidimensional space will be. Are both parallel interpretations of the same reality? Can one entirely account for the other? If so, which is the more general view? Do they have a zone of intersection and zones not in common? Considering the importance of this question and the new perspectives it opens up, I have added Part IX, which I consider in the nature of an introduction to the subject and to its many promising possibilities in the study of mental phenomena. It must be clear, however, that I do not intend here to make a complete presentation of a view of the mind in terms of multidimensional space. But I hope that what is presented is sufficient to show that the points raised on this subject at various stages in the book may lead, if developed, to a further, stimulating increase in our knowledge.

I.M.B.

June 1974

Acknowledgments

If I were to acknowledge my indebtedness to all those who, in one way or another, have contributed to my writing this book, the list would be too long (for the reader). I wish, however, to express my heartfelt gratitude to the following:

The late Professor R. Courant. He showed a considerable interest and appreciation of my use of the concept of multidimensional space in connection with mental phenomena. In the midst of his important work he found time to hear me and to invite some of his colleagues for a discussion of my ideas.

The late Dr Paul Federn. He opened the discussion of a paper on space and mind which I read before the New York Psycho-analytic Society and made positive and encouraging comments which helped to build my confidence in my work.

The late Dr Ernest Jones. He was very generous about my efforts towards 'reformulating our knowledge' of psycho-analysis, as he put it. Encouragement from him was important in my searches.

Professor Gerold Stahl. He was my first teacher in symbolic logic and encouraged and helped me when I wrote my initial paper on the subject.

The late Bertrand Russell. He played an important role for me, not only on account of his writings but also because, in a short note, he expressed an appreciation for my work. (The paper which he saw is, with some changes and additions, contained in Chapter 3.)

Dr Massimo Pallotta and Professor M. Lippi, who kindly revised the typescript of Chapter 2 and of Parts IV and V (all of which I subsequently enlarged).

Dr Jacques Mehler, who made some valuable criticisms of the methodology implicit in Parts IV and V.

Professor J.O. Urmson, who very kindly read the manuscript at an advanced stage and made some significant remarks that led me to clarify certain expressions which were susceptible of ambiguous interpretations. This, in its turn, stimulated me to develop further some of the basic arguments of the book.

Dr R.E. Money-Kyrle. He has been extremely generous to me, with his time, his thoughtful and stimulating comments, his friendly encouragement and in various other ways.

Anthony Maclean. He carefully revised the text so as to ensure

that my writing remained within the limits of respectable English, while at the same time leaving room for my stylistic idiosyncrasies. His help, however, has gone beyond that, for had it not been for his presence, I should probably have procrastinated and this work would have taken much longer to reach completion.

Simon Marks. He has carefully gone over the text not only to ensure a satisfactory editorial presentation, but also to discuss with me any expression or phrase which seemed obscure. Thanks to him I have succeeded in eliminating various imprecisions and small defects.

The text has been improved with the help of the persons mentioned. I must, however, point out that the responsibility for the ideas put forward and for any errors or faults which may have slipped in is entirely mine.

In a book of this kind it has been necessary to quote from time to time from a number of authors. I am grateful to all copyright holders; full references for works quoted are given in the Bibliography. For permission to quote from Freud I am grateful to Sigmund Freud Copyrights Ltd, Hogarth Press and the Institute of Psychoanalysis.

Finally I wish to thank Heather Kirk-Duncan who has efficiently typed most of the book, patiently transforming a hieroglyphic into a correct and presentable manuscript; and Helen Chroscicki who has typed Parts VIII and IX as well as various additions to the text.

I.M.B.



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PART ONE

Introduction

. . . the other view, which held that the psychical is unconscious in itself, enabled psychology to take its place as a natural science like any other. The processes with which it is concerned are in themselves just as unknowable as those dealt with by other sciences, by chemistry or physics, for example; but it is possible to establish the laws which they obey and to follow their mutual relations and interdependences unbroken over long stretches – in short, to arrive at what is described as an 'understanding' of the field of natural phenomena in question.

This cannot be effected without framing fresh hypotheses and creating fresh concepts; but these are not to be despised as evidence of embarrassment on our part but deserve on the contrary to be appreciated as an enrichment of science. They can lay claim to the same value as approximations that belong to the corresponding intellectual scaffolding found in other natural sciences, *and we look forward to their being modified, corrected and more precisely determined as further experience is accumulated and sifted.* So too it will be entirely in accordance with our expectations if the basic concepts and principles of the new science (instinct, nervous energy, etc. . . .) remain for a considerable time no less indeterminate than those of the older sciences (force, mass, attraction, etc. . .). (Freud, *An Outline of Psycho-analysis*, 1940, pp. 158-9, my italics)

1. *Scope, Outline and Meaning of this Book*

1. The difficulties of present-day psycho-analytic theory

In the course of its development psycho-analysis has outgrown to a considerable extent its own theory and finds itself in a situation comparable to that of an adolescent who has outgrown his clothes and feels restricted, hampered in his movements and uncomfortable. The finding of significant new facts – and I do not mean by this just simple elaborations of already well-established discoveries – has become increasingly difficult on account of the lack of an appropriate frame of reference against which new facts can be seen. The result is that much of the inexhaustible wealth offered daily by clinical reality is simply not seen because it does not fit in with the theories in use. If this situation continues indefinitely psycho-analysis, as a science and as a technique for helping patients, runs the risk of falling into a circular and sterile pursuit of an ever-escaping psychical reality; it will become like an animal attempting to catch its own tail. As a result of this, the image of the dogmatic, self-satisfied analyst who is closed to and afraid of new developments is becoming increasingly frequent, and one is reminded of the futile subtleties of some medieval theologians or of some Talmudic interpreters. But one is also reminded, and this is distressing, of the evolution of some animal species which in the course of time exaggerated the development of certain characteristics to the detriment of their survival; this, for instance, was the case with some wild boars whose tusks became so twisted that they finally made eating an impossibility.

There is no doubt that psycho-analytic theory needs reformulating and we are witnessing at present various attempts at doing so.

It seems that it is necessary first to try to understand *where* psycho-analytic theory is insufficient. In order to do this satisfactorily one must have some clear ideas on a fundamental question: the relation between theory on the one hand, and facts and fact-finding on the other.

I believe it is accurate to say that at the present moment a large proportion of analysts tend to avoid theorising in order to concentrate, so one frequently hears, on clinical facts. Such an attitude seems ingenuous and those who adopt it seem to be unaware

that in fact they are living on borrowed income: somebody else's income. For they do not seem to realise that the facts they discover are precisely, and no more than, those which the theories to which they subscribe enable and allow them to find. The examples in favour of this assertion could be multiplied at will, and I shall only mention one or two. When the so-called castration complex was found, a great deal of the clinical material was interpreted in terms of it, and nothing was seen in terms of, say, introjection or envy of the breast. It would be unwarranted to assume that the patients seen in that period did not show any signs corresponding to these later concepts: only that such signs were not noticed, for the simple reason that the corresponding concepts were not yet available. In a similar way, the type of interpretation we nowadays give may be an accurate reflection of the clinical material at hand but not necessarily a *complete* reflection of it. We do not know how many things pass unnoticed by us in the data offered by our patients, simply because we do not have the frames of reference which would enable us to see such things.

When (for example) we hear people giving interpretations in terms of relations with the internal object, or of the role of container played by the analyst, or of the envy of the breast, as also represented by the analyst, one frequently gets the impression that such interpretations are correct and conform with the clinical evidence. At the same time, when observing such commendable therapeutic activity sometimes one cannot avoid feeling that the therapist is convinced that his interpretations cover the totality of what is happening to the patient and that he is implicitly conveying his own certainty that there is no more to be known about it. It is in cases of this type that one can see the intimate link existing between the frame of reference employed and fact-finding. Interpretations of the type just mentioned are based, as can easily be seen, on a three-dimensional analogy; if instead of this analogy we were to use, say, a five- or six-dimensional analogy, the whole material presented by the patient would be seen in a quite different light. Various interrelations so far invisible would become evident and one would have a new understanding of the patient. The 'internal object' itself would then become something corresponding more intimately to actual psychical reality, for it would then be possible to realise that something which at a lower dimension is experienced as a separate object, becomes, at a higher dimension, a constituent of the whole. One would, in this case, be more able to approach the proper formulation of a question which has become urgent in recent times: what are the respective roles of the unity of the individual and of internal objects? On the one hand, we cannot avoid feeling that each individual is *one* and is something that confronts other individuals; on the other, we cannot reject the evidence in favour of the internal object. It must be recognised that present-day psycho-analysis has no satisfactory answer to this question.

To return to the internal object and the container, we may now consider an alternative way of understanding the data given by our patients. The notion of multidimensional space (which is the concept referred to in the alternative mentioned above) is not studied exhaustively in this book and only some initial approach to it is made. Instead, the relation between space-time and spacelessness-timelessness, is the object of a more detailed study. Now, if the material offered by patients is systematically studied in this light, many as yet unnoticed facts become apparent and we are then able to understand and help our patients better. We may, for instance, find that the same material shows signs of several levels (in the sense studied in Part IV) in action simultaneously. At a more superficial level we may find the relationship between separate persons, whereas at an intermediate level, instead, the persons are 'lived' as objects which may be inside, or inside which one may be. This is the level of envy, of projective identification and containers. At a still deeper level the distinction between persons or between objects begins to lose sense in the same proportion as spatio-temporal notions begin to fade away. Correspondingly, the concept of aggression, after having passed through levels of infinite magnitude, begins to recede into the background. The fundamental unity of subject and object makes itself increasingly felt, until a moment comes when speaking of projective identification no longer has any sense; we are at the level of the *basic matrix of projection and introjection.* The notion of envy is, at this level, no longer pertinent.

The above is only a summary and a schematic description of an alternative way of looking at given material which had first been interpreted in terms of current analytical views. It would be possible to give many more clinical examples which would show in detail how, starting from the associations of the patient, one may reach different types and different degrees of understanding, and hence of therapeutic help, according to the type of approach. The type of approach, in turn, depends on the frame of reference used in the study of reality; or, as is usually said, it depends on the theory in the light of which reality is seen.

For this reason it becomes imperative to reach a clear stand on the question of the relationship between facts and theory.

Clinical facts, frame of reference and theory. At this point we must ask: what actually is a fact, whether clinical or otherwise? The position which assumes that facts are things in themselves, independent from the way they are observed, is frequently found among clinicians and is the origin of much superficial theorising, of which people are not aware. Let us consider a very simple fact: the time at which an aeroplane arrives at a given airport. It may be, for instance, 6.30, and this appears to be quite precise. But if the country where the plane arrives has adopted summer time it will be 7.30. It will,

then, be equally true to say that the plane arrives at 6.30 or 7.30. Now if the country from which it started is very far off, the time of arrival expressed in terms of hours may vary from 6.30 to 19.30, according to which country's time is chosen to express the arrival in question.

We may also adopt another convention and correlate the time of arrival with the rising of the sun at the place where it arrives. In this case the plane will arrive before sunrise, at sunrise or after sunrise, according to the time of the year. If, instead, we prefer to change the place in relation to whose sunrise we measure the arrival, we may then arrange things in such a way that every day of the year the plane will arrive exactly at sunrise, only the sunrise of each day will be that of a different place from the sunrise of all other days of the year.

If instead of adopting a 24-hour day, we adopted a different convention, we might easily arrange things so that the plane arrives at a given day or the next, according to the convention preferred.

If we now consider the word 'aeroplane' we soon realise that it means a complex system of relations, say between wings, fuselage, methods of propulsion, etc. A bird has wings but it is not a plane. Cars have engines, like planes, but they are not planes. Wing, fuselage, engines, etc. are, themselves, sets of relations.

> And so we finally arrive at the conclusion that *all our knowledge of the world is ultimately a knowledge of relations* to which that which we call world conforms (in a greater or lesser degree, according to how appropriate the relations chosen are). The true reality, the noumeno, of the world is unknown to us. A fact is, therefore . . . a relation between two events; and an event is itself another relation, so that in the end the only things we may discover are relations, relations between relations, relations between relations between relations, and so on. We could define these as events. And I imagine that it is because of this that some contemporary philosophers and mathematicians have come to define the world as composed only of events, which do not happen *to* matter, nor to anything else, but simply happen. . .
>
> A fact is something that is always defined against a frame of reference, and this frame itself is a system of relations. The same underlying 'reality' may be described against various frames or systems or relations, some of which may be more appropriate to it and belong to a more general order, which will permit us to make more accurate predictions and a greater number of them. (Matte Blanco, 1954, p. xxxi)

The position just outlined is that generally adopted by philosophers of science, and unfortunately frequently ignored by researchers, who sometimes do not seem to be aware that they are never describing *facts in themselves*, because such things do not exist. Perhaps it is useful to give two significant quotations to clarify the meaning of the concept underlying the word 'theory'. Von Mises (quoted by Szasz, 1959) writes:

> Since the time of Ernst Mach, natural scientists have known that the explanation or the theory of a group of phenomena is only a description of the facts at a higher level.

On the other hand, Braithwaite (1953, pp. 367-8) says:

> Nature does not provide separately both facts and laws; our statements of laws are a way of describing observed facts and of predicting facts at present unobserved. The form of a statement of a scientific hypothesis, and its use to express a general proposition, is a human device; what is due to Nature are the observable facts which refute or fail to refute the scientific hypothesis.
>
> . . . The function of mathematics in science has been shown to be, not that of admitting only hypotheses of a pre-ordained form, but that of providing a variety of methods for arranging hypotheses in a system; knowledge of new branches of mathematics opens up new possibilities for the construction of such systems.

As can be seen, this author speaks of hypothesis, system and law. In psycho-analysis it is common to speak of theory. Personally, I prefer to employ the term 'frame of reference', rather than 'theory', because the latter is historically loaded with a collection of meanings which are better avoided. I am referring to the fact that the word theory frequently evokes the concept of guesses, which may be more or less warranted, more or less happy, or be accurate; or elaborations which may be rather distant from our initial observations and which may or may not turn out to be true. In contrast the expression 'frame of reference' has a more restricted meaning: that of a system of relations into which our observation of reality directly fits. In other words, it is more immediate, and aims at being more directly in contact with the reality under study. It is more directly at the service of this reality, ready to be changed as soon as new observations make us aware that the frame is not capable of describing them satisfactorily. But such a distinction between theory and frame of reference is ultimately a matter of convention and we may agree to employ both terms as synonymous. What seems to be important to keep in mind is that sound scientific research aims at creating frames of reference which are, so to speak, directly suggested by our intellectual contact with reality and which are modified as fresh contacts suggest changes in the frame which is being used.

Psycho-analytic theories or basic concepts have become manifestly insufficient. Psycho-analysis has been developed within the frame of three closely interconnected concepts: instinct, energy and space. The first refers to the biological nature of man, whereas the latter two establish some contact between the concepts of mind and matter, since both of them are also fundamental in the study of inanimate nature. As Freud himself has remarked, this 'intellectual

scaffolding' may be 'modified, corrected and more precisely determined as further experience is accumulated and sifted'. It seems that this is precisely the present case of all three. The concept of instinct has been the object of intensive studies among biologists, ethologists and psychologists, and though its essential features do not seem to have changed since the times of William James, an enormous amount of information has been gathered in the course of time which is of great relevance to psycho-analysis as a conception and a technique. Much work needs to be done in order to introduce this relevant information into psycho-analytical thinking and this task has already been started by various researchers, foremost among whom is Bowlby.

The concept of energy is the basis of the dynamic and economic points of view in psycho-analysis as generally accepted. The concept of space is the background to the topographical and structural points of view and to the notion of object. As this assertion may be questioned it deserves some comment. Freud preferred to employ the term 'topographic', but he explicitly treated 'structural' as synonymous to it, as we shall see in Part III. 'Topographic' clearly refers to space, as do expressions such as 'depth psychology', 'deep unconscious', 'surface of the mental apparatus', 'barrier', 'keeping *away* from consciousness', 'return of the repressed', 'projection', 'introjection', 'internal object', 'internalisation', 'externalisation', 'external object', 'container', 'blowing up', 'tearing into bits', 'reparation', 'deviation of energy' (sublimation), 'displacement', 'turning against the self', and various others. Some, instead, question the accuracy of the view that the so-called structural conception is a topographical one. The fact is, however, that it employs the same spatial *metaphor* as all the expressions just mentioned. This can easily be seen if one considers that Freud spoke of 'regions or provinces' of the mind, and that this is a topographical comparison. Freud also made a diagram in which at least some of the relations between the three psychical instances are studied in terms of the comparison with space.

All the same, the problem is that all the spatial comparisons so far usually employed in psycho-analysis are appropriate to describe physical phenomena and are insufficient to describe mental phenomena. In fact all such comparisons are based on the three-dimensional analogy and it is extremely improbable that psychical phenomena can be described in terms of only three dimensions. Yet, the psycho-analytic literature is most strangely silent about this fundamental premiss of psycho-analysis. Freud gives no hints or suggestions on this point, though it is obvious that he thought about it until the end of his life. His last but one published phrases on psycho-analysis read as follows (Freud, 1941, p. 300):

> Space may be the projection of the extension of the psychical apparatus. No other derivation is probable. Instead of Kant's *a priori* determinants of

our psychical apparatus. Psyche is extended; knows nothing about it.

I know of no evidence which shows that he ever thought of applying to the mind spatial concepts not borrowed from those employed in the study of 'material space' or three-dimensional space. And the recent abundant literature about object-relations, internal objects, etc. is, to my knowledge, utterly unaware, in its essentials, of the magnitude of the problem and its important consequences in the field of clinical work. There is a great need of clarification in this large subject.

Exactly parallel considerations can be applied to the use of the notion of energy. It has remained extremely vague and nothing essential has been added to Freud's initial intuitions. As in the case of space, the notion of energy seems to be of great importance in understanding the mind. But the use made of it by those researching into the structural conception is so unsatisfactory that it is no wonder that many people keep away from it altogether. Terms such as 'desexualisation', 'desagressivisation', 'mental energies', 'the ego's autonomous sources of energy', 'reservoirs of energy' and the like are extremely vague. If no serious attempt is made to deepen our understanding of the basic concept of psychical energy as, for instance, the question of the possibility or impossibility of measuring it, and its relation (similarities, differences or identity) to other forms of energy, then no real progress can be made.

So, in the end, we find that all three basic concepts of psycho-analysis are in need of revision, because they are unsatisfactory in their present form. The trouble is, however, that much effort is devoted to applying these concepts and to describing clinical reality with their help, and not sufficient time to studying the concepts themselves. It seems as though many implicitly believe that this is the way to be 'clinical'.

In the meantime, it can in all fairness be said that psycho-analysis has neglected to a considerable extent its initial purpose of exploring the psychology of the unconscious, and of that mysterious world where everything is so different from what we see in conscious life; for it is quite evident that both the structural and the internal object approaches describe the mind in such terms that the characteristics of the system unconscious either become unnecessary or are rendered trivial. In the course of its development psycho-analysis has become less psycho-analytic in the sense that, though it continues to deal with so-called unconscious *contents*, it tends to treat them as though they were ruled by the same laws that are seen in consciousness and applied in the study of all the other sciences. Psycho-analysis has lost its most distinctive characteristics. If this were the result of observations which led one to correct the initial formulations, then it would be a sign of progress. But the fact is that this is the result of neglecting the essential theme of psycho-analysis,

probably because of its extreme difficulty. If we were to describe this process in terms of early psycho-analytic terminology I would say that the discovery of the unconscious has been repressed in psycho-analytical thinking and subsequently replaced by neatly constructed rationalisations, which can be described in terms of material space and energy but which hide the underlying reality of the unconscious as 'the true psychical reality'. To put it paradoxically, *psycho-analysis has wandered away from itself.* In the same vein Freud wrote

> of the embarrassment that still comes over us when, accustomed as we are to the atmosphere of the underworld, we move in the more superficial, higher strata of the mental apparatus. (Freud, 1933, p.68)

Now, it seems to me, the situation has been overturned. Not only do those who cultivate the so-called ego psychology seem to feel much more at home in 'the higher strata', but also those who deal with 'deep material', such as the early relationship of the child to his mother, usually treat this material as though it were ruled by the laws of the preconscious, even though at times some passing reference is made to the contrary.

2. The scope and meaning of this book

It is as a result of prolonged reflection about these subjects, spread over many years, that I have gradually come to develop the ideas which I am putting forward here. It may briefly be said that the essential meaning of the present work derives from an effort to think systematically about psychical reality in terms of the relationship unconscious-conscious (to employ Freud's initial formulation), or symmetrical-asymmetrical (if we use this conceptual duality, which, as will be seen, offers some advantages over the former). When I say 'in terms of the relationship conscious-unconscious' I am not referring only, or mainly, to the quality of being conscious or not, but to the two contrasting modes of being visible in this relationship. In other words, the present approach aims at taking the meaning of the characteristics of the system unconscious described by Freud very seriously and following this meaning to its logical conclusion; it also aims at studying the 'omnipresence' of this system in every mental manifestation as well as its infinite interrelations with the so-called conscious mode of being.

In order to help the reader follow the arguments put forward here, which may initially present some difficulty, it seems advisable to give a brief perspective of the path taken. If we read what Freud has written on the subject of the characteristics of the system unconscious we soon realise that he was explicit in affirming, especially towards the end of his life, that the laws of Aristotelian logic were not followed or respected in the system unconscious or in the id:

> The logical laws of thought do not apply in the id, and this is true above all of the law of contradiction. (Freud, 1933, p.73)

To put it in another way, he described or referred to a deficiency, to something missing, and not to a different type of logic. It seems that this idea has been prevalent in the relevant literature, even if not necessarily in an explicit manner. This deficiency may be considered in two ways: either as leading to a complete disorder, or as resulting in a new order. When Freud writes (1933, p. 73) that

> we approach the id with analogies: we call it a chaos, a cauldron full of seething excitations

he seems to imply the first of the two alternatives. But when he writes (1900, p. 507) that

> the dream-work is not simply more careless, more irrational, more forgetful and more incomplete than waking thought; it is completely different from it qualitatively and for that reason not immediately comparable with it

he does not appear to be thinking of the second. The fact remains that, as far as I am aware, he never arrived at a description, in precise logical terms, of this different order. It is at this exact point that the developments suggested fit in. In 1956 (Matte Blanco, 1959) I proposed that the 'special characteristics of the system unconscious' described by Freud are various expressions of a type of logic which I then formulated in terms of two principles; I called these *the principle of generalisation and the principle of symmetry*. With their help it was possible to see the intimate connection between these five characteristics described by Freud as conceptually independent, even though in fact they frequently appear simultaneously. With the help of the two logical principles all five were seen as an intimate unity, the expression of a type of logic which could be defined precisely, even if such a definition showed a peculiar and characteristic breadth of approach to reality. Moreover, the systematic application of the two principles results in a type of thinking that differs strikingly from that usually employed in science, which follows so-called Aristotelian logic.

It then became possible to verify that the logic of the system unconscious or logic of the id – which we may also call *symmetrical logic*, in order to avoid the ambiguities connected with the notion of system unconscious and also with that of id – is quite consistent and always conforms to the two principles in question. In a series of studies I have focused on various manifestations of this logic in a wide variety of phenomena. Among these I may mention: schizophrenic thinking, feeling and *erleben* (1959, 1966, 1967*a*); the unconscious contact with external reality (1960); Jung's syncronicity

as a principle of non-casual relations (1962); the contrast between neuroses and psychoses (1964); various applications to psychiatric problems such as mental confusion (1965); non-verbal communication (1962*a*); the formulation of the basic concepts of psychoanalysis (1968); the question of interpretation (1968*a*); introjection and the problems arising in connection with the notion of mental objects, as well as in the mechanisms of projection and projective identification (1960*a*, 1970); the question of the 'nature' of emotion and its relation to thinking and to consciousness and the unconscious (1967). From these studies as a whole, I believe it can safely be said that several general conclusions emerge, some of which have a bearing on the present study, in the sense that they represent a general background against which this study's subject matter is projected:

(1) Clinical observation is sharpened if we make our descriptions with the help of logico-mathematical tools (which are simple and easy to use). It then becomes possible to discover a variety of clinical facts which would otherwise have escaped us.

(2) The relationship between the timelessness of the unconscious and the 'timeness' of those aspects of our self which are oriented towards the external world can be explored further. As can be seen (Matte Blanco, 1959), timelessness is a necessary consequence of the principle of symmetry, because, if asymmetrical relations are not available, there can be no time in the physico-mathematical sense of the word. It must be added that, for the same reason, neither can there be space; the consideration of this question has led me to propose (Matte Blanco, 1968) adding spacelessness as another of the characteristics of the system unconscious. This proposal only amounts to adding explicitly another aspect of so-called symmetrical logic. Freud had considered this idea from more than one angle, though not as part of a logical system (see, especially, Freud, 1933, p. 74 and 1940, p. 300). If we remember that both in the physical world and in the world of the mind, time and space are inextricably linked, the convenience of such a proposal is evident.

As already implied in this Introduction, the characteristics in question, described by Freud in his paper 'The unconscious' (1915) which ranks, in his own opinion (Jones, 1956, p. 34), among his most important contributions, have hitherto played a comparatively modest role in psycho-analytical development. It is true that such an assertion would not be completely accurate if it referred to clinical work, where displacement, condensation and replacement of external by psychical reality are habitually employed. But if we consider psycho-analytical conceptions and theoretical developments, then it is certainly accurate to say that they have not yet been exploited to their full extent. I will give as an illustration the fact that various recent significant psycho-analytical findings on schizophrenia are described in terms which largely correspond to the *logic* of conscious

thinking. I am, for example, unable to find the role that timelessness plays in Melanie Klein's schizo-paranoid position. Furthermore, the question must be raised of how we can speak of putting inside or introjecting an object, when the unconscious does not know 'inside' or 'outside' and does not know objects. And all of this flows from this particular aspect of Freud's work.

Conversely, if we start from these characteristics, reformulated in logical terms, we can give the problems the full attention they deserve (see, especially, Matte Blanco, 1970). It soon becomes evident that we are always, in a given mental product, confronted by a mixture of the logic of the unconscious with that of the preconscious and consciousness.

The proportion between both components of the mixture differs greatly from one case to another, so that in the end there are innumerable types of cases. In order to simplify matters I have proposed (Matte Blanco, 1968*a*), following some remarks made by Freud in 'The unconscious' (1915), to distinguish roughly three levels, especially with regard to the role played by spaceness-timeness and spacelessness-timelessness; though the same distinction also applies, of course, to all the other characteristics.

(3) The most important general conclusion that emerges from these studies is that psychical life can be viewed as a perpetual dynamic interaction – in terms of tension, co-operation, or even union – between two fundamental types of being which exist within the unity of every man: that of the 'structural' id (or unrepressed unconscious or system unconscious or symmetrical being) which becomes understandable with the help of the principle of symmetry; and that visible in conscious thinking, which can roughly be comprehended in Aristotelian logic. This formulation is an expression of Freud's discoveries in terms of the ideas mentioned; it has the advantage of enabling us, as I believe, to draw hitherto unexploited riches from Freud and from clinical reality; it also stimulates new developments in psycho-analytical theoretical thinking. It will be noted that this way of looking at the question – which is, to a great extent, related to Freud's formulation of the mind in terms of the unconscious and the conscious ego – is the result of studying clinical reality with the help of logical tools. It may be objected that it represents a withdrawal from Freud's later conception of the threefold structure. In my opinion it should rather be viewed as an attempt to preserve something which was valuable in the former conception and which has somewhat receded into the background in the later conception.

All the above constitutes, as I have already remarked, the background against which the present study is projected, but the subject extends in other directions which fit in as integral parts of the whole I have just outlined (see Section 3, below).

The relation of the present approach to the basic concepts of instinct, energy and space. It will have been seen that in the above description of the meaning and scope of this book no systematic mention is made of the three 'basic concepts and principles', to take Freud's expression, upon which psycho-analysis is based. This fact raises the question of the relationship between this approach and such principles. It can be said, in this sense, that the basic questions of energy and space are not studied in themselves here, at least not in an exhaustive way; instead, an attempt is made to develop the basis of our psycho-analytic knowledge, starting from the characteristics of the system unconscious, reformulated in logical terms. This is neither a frontal attack on the questions of energy and space nor a retreat from such questions. It is indeed, a way of coming to them from another angle; precisely that of the unconscious or symmetrical mode of being (and its relation to the asymmetrical mode), which may in the end lead to much more fruitful developments and to a clarification of many of the problems related to these basic questions.

In fact, an attentive reader will become aware that the formulation of the question of the measurability of unconscious processes touches on the question of the measurability of psychical energic processes, though from an angle which is quite different from that of a direct approach to the question of energy. On the other hand, the consideration of the symmetrical-asymmetrical duality leads us to the problem of space-time, a continuous study of which is made throughout the book. This problem, however, can be tackled directly, for instance with regard to the notion of dimension. This is considered in Part IX.

It may legitimately be said that after the study of the various subjects of the book our knowledge of energy and space-spacelessness, as far as it concerns the mind, emerges enlarged and deepened.

A 'lateral' attack on these questions may, at present, turn out to be more fruitful than a frontal assault.

As for the basic notion of instinct, it is taken for granted here and only a few comments are made about its relation to the subjects studied in the book.

The relation to Freud's conception. At first sight the present study may give the impression of something considerably different from Freud's conception. In my opinion it remains well within this conception, only it is formulated in terms which permit of further developments in accordance with the accumulated clinical experience of many psycho-analytical observations made by many analysts over a long period. It can be shown that the developments put forward here usually start from Freud, and that an effort is made to understand his thinking and, following his formulated opinion and

wish (as shown in the quotation at the beginning of this Introduction), to modify them and develop them as the need to do so arises from the very nature of things. The numerous quotations from Freud throughout the book and the efforts made to tackle and draw conclusions from the various thoughts expressed by him, confirm that on no occasion have I taken lightly the thinking of Freud; on the contrary, I have made it the object of thoughtful study. This inevitably leads, as Freud hoped, to developments and modifications of his ideas. The spirit remains the same.

I have thought it necessary to make these qualifications, first, to give a better perspective of this work as I see it and, secondly, to avoid any misunderstanding.

A further reflection may be relevant. I believe it can truly be said that the present work lives to a great extent in the atmosphere of the early Freud, that of the heroic saga of the *Interpretation of Dreams*, as well as in the atmosphere of the *Outline of Psycho-analysis*. Both works, at the beginning and at the end of his creation, represent two fascinating outbursts of his creativity. As Strachey rightly points out (1964, p. 143), there are in this latter book 'hints at entirely new developments'. If one is careful not to take too literally what I am just saying, one might think of these two books as supreme expressions of Freud the creator, in contrast to Freud the organiser of his discovery, seen more prominently in other of his works. I believe this is, for instance, the case of *The Ego and the Id*, despite the protests that this assertion may provoke.

I hope the present approach may not only be of interest to analysts but offer at the same time the possibility of new forms of dialogue between psycho-analysis and philosophy, mathematics, logic, moral philosophy, anthropology and sociology. All these disciplines have contributed to its shaping; perhaps the formulation put forward here, in its turn, will pay back to them, in a psychological form, something of its debt to them.

At the same time psycho-analysis could become the link between the inevitable regimentation of modern civilisation and the unforeseeable freshness of emotion. There is no other discipline which is both scientific and moves in the realm of emotion and, hence, of art and also, in part, of ethics, sociology and politics: in short, of the humanistic side of man, because it deals simultaneously with the duality unconscious-conscious or symmetrical-asymmetrical.

3. Brief summary of contents

I shall now make a summary review of the contents of this book so that the reader can have a panoramic view of the course followed:

Part II. As the title shows, this deals with the notions which are indispensable (to those who are not conversant with them) to

following the arguments of the book. The first chapter deals with the simple mathematical notions required. The second is an enlarged version of one of my first two papers which initiated this line of thought. It is hoped that with a knowledge of this chapter in hand, the reader can follow the rest of the book without having to read the other related papers.

Part III. This constitutes the first published presentation and development of my studies on a subject which has great relevance to the rest of the book. It also represents the link and transition between current psycho-analytic thinking and the present approach. It is hoped that reading this part will make things more understandable; it will explain how the present approach starts from current psycho-analytic thinking and why it came to be required in the course of the evaluation of the problems which confronted this type of thinking.

It starts with Freud's concept of the unrepressed unconscious, the study of its meaning, scope and difficulties. It goes on to consider the initial conception of Freud in terms of consciousness and the unconscious and its reformulation in terms of symmetrical and asymmetrical modes of being. At the end a study is made of the threefold conception in the light of these ideas and some proposals are made for a reformulation of the threefold conception. An attempt is made to introduce some order to our ideas, and the advantages and the disadvantages of the introduction of the threefold conception are discussed, as well as the difficult problems that this introduction has led to. Also, among other things, a study is made of what I have proposed calling the function of 'translating' or 'unfolding', which is, for the unrepressed unconscious, the counterpart of what lifting of repression is for the 'repressed unconscious'.

Part IV. This deals with the unrepressed unconscious seen as infinite sets. This is a modified version of a paper presented in 1970 to the British Psycho-analytic Society and it is the first published study of my personal work on this subject. With the help of our logico-mathematical tools a variety of clinical facts and concepts are outlined. Among these, the following are proposed:

(1) The interpretation of the principle of symmetry in terms of infinite sets, starting from the observed fact that the unconscious establishes the identity between the whole and the part. This fact, in its turn, can be viewed as a consequence of the principle of symmetry, all of which shows the conformity of the logical formulation proposed with actual psychical reality. The strange identity between the whole and the part, so much in contrast with Aristotelian logic, led me to apply Dedekind's definition of the infinite set as the set in which the whole and the proper part have the same cardinal number. If we interpret the identity between the

whole and the part as meaning that they have the same cardinal number, we must conclude that whenever we come across such identity, the unconscious is dealing with or considering an infinite set.

(2) The notion that the deep unrepressed unconscious deals with infinite sets and only with them. This throws a new light on the concepts of displacement, symbol, sublimation and a variety of other basic psycho-analytical conceptions.

(3) The proposal of a distinction between extensive and intensive infinite sets, which permits differentiation between various types of mental manifestations.

(4) The notion of mental functions which have a maximum and a minimum, both of which are of infinite value. In other words, what I have called the law of either positive or negative infinite sets. This touches on the question of the absence of zero in the unconscious and makes it possible to view Freud's intuition regarding the absence of negation in the unconscious under a new light.

(5) The notion that the unconscious deals with infinite sets that have not only the power of the denumerable but also that of the continuum. In simple, though probably inexact terms, it deals with infinites of infinites. This seems to be of importance in the understanding of the mind. The consequences and developments of such notions are unforeseeable now, but certainly capable of opening up vast possibilities.

(6) The notion that the magnitudes of emotion are a function of the level of depth, which, in this respect, is understood not as stages of development but as the proportion between symmetrical and asymmetrical thinking. This is of theoretical as well as of practical value in analytical practice. Put in simple terms, the more there is, in a given manifestation, of the deep unconscious, the greater will be the magnitude (in the mathematical sense of the word) of emotion: in the deep levels there will be an infinite value of the same magnitude. This formulation, as I see it, may be the starting-point of new developments regarding the magnitude of the affects, of the so-called cathexes, viewed from a different angle.

(7) The notion of the coexistence of the various levels in a given person at a given moment. I think this has been known since the beginnings of psycho-analysis, but it is put forward here in a different way; among other things, it means that the same emotion simultaneously has different values of the same magnitude. This is puzzling, though quite in keeping with the arguments developed here. Obviously, its implications have to be further explored.

(8) The notion that towards the deepest levels, where there is 'pure symmetry', hence, no space-time notions (that is, the level of *being*, in contrast to the level of *happening*), there can be no aggression in the sense of destruction. I believe this view throws light on the concept of the death instinct, which was perhaps one of

Freud's deepest intuitions but which he presented in an obscure manner, very much open to criticism, because he did not employ appropriate technical tools. This approach leads to another way of looking at the problem of evil in man.

(9) The notion that analytic therapy can be viewed as divesting a given situation of the characteristics of an infinite set. There is much to develop here.

Part V. This deals with the infinite sets and the question of the measurement of unconscious processes. It starts with a review of the classical notions about measurement; then it discusses in general terms the possibility of measuring psychical phenomena and the various questions and difficulties connected with this. Finally, it goes on to study the possibility of measuring unconscious processes. It arrives at the paradoxical conclusion that unconscious psychological events are not intrinsically immeasurable but, in contrast to physical events, are susceptible of infinite measurement at the point in which the physical event is susceptible of only one measurement. It also discusses the merging of the conceptually measurable into the conceptually non-measurable and refers to the consequences of such a view.

Part VI. This deals with the nature of emotion. It is very difficult to give a satisfactory summary perspective of the various problems discussed in this Part. It starts with a phenomenological-psycho-analytical approach to the concept of emotion. It shows that emotion is a composite psycho-physical phenomenon and that in its psychological aspects it consists of two basic components: sensation-feeling and establishment of relations. It studies each of them in its multiple aspects; among other things, the passage from sensation-feeling to image and perception is studied; as well as introspection and time, attention, 'pure sensation', and the relation of these phenomena to space-time.

Regarding the establishment of relations aspect of emotion, a detailed study of its various facets leads to the conclusion that the most specific characteristic of emotion is the establishment of symmetrical relations. A comparison is made between the view of emotion proposed here and those put forward in the current literature. Among other things the similarities and differences between the present view of emotion and Sartre's view are discussed (see Appendix).

The question of the measurability of emotion is discussed in two chapters. It is shown that emotion bears several different relations to the concepts of magnitude, quantity and measurement, and these various relations are discussed in detail. The relation between sensation-feeling, imagination and perception, and their respective roles, are discussed in detail.

Regarding the 'establishment of relations component' of emotion, this latter is seen as infinite sets, dealing with classes having infinite values of the various aspects of the propositional functions defining classes. This leads to the conception of emotion as having in itself various infinites. On the other hand, emotion, as the unconscious, can be seen both as measurable and as non-measurable, according to the vantage point from which it is observed. In the latter of the two cases, non-measurable emotion can be considered as the 'mother of the measurable'.

A special chapter is devoted to the development of the concept of the translating function. This gives the opportunity of a further study of asymmetry as the 'restraining condition' required in scientific hypotheses.

The relationship between instinct and the symmetrical-asymmetrical polarity is studied and the conclusion is reached that instinct makes what I call a 'lateral insertion' into mental life.

A detailed study of the relationship between symmetrical and asymmetrical aspects in man is made and the consequences that such relations entail for the understanding of mental life are discussed. The concept of 'boundary' is considered here and this leads to a generalisation of the concept of repression.

The significant question of the 'pouring' of feeling into thinking or, alternatively expressed, the 'extraction' of thinking from feeling is studied in the context of the obscure relation between thinking and feeling. This, in its turn, leads us to consider the question of the 'light' of consciousness and the 'darkness' of the unconscious. Here we come across the notion of the 'induction situation'.

The question of 'making the unconscious conscious' and of 'where id was there ego shall be' is discussed under the light of the considerations developed in this Part.

The last chapter of this Part is devoted to discussing the question of the place of emotion in the psycho-analytical conception. The apparently surprising conclusion is reached that if we formulate the psycho-analytical conception of the mind in terms of the interaction between the symmetrical and asymmetrical modes of being (from which follows the notion of the unconscious), then the concept of emotion, together with that of its relationship with thinking, comes to coincide entirely with the psycho-analytical conception of the mind. This formulation, it is felt, helps in the understanding of various problems concerned with the so-called psycho-analytic theory of affects. At the same time it provides a solid ground for the integration of psycho-analysis with psychology.

Part VII. This represents an attempt at formulating a general theory of the bipolarity unconscious-conscious or symmetrical-asymmetrical. In my opinion a most significant aspect of the study made in this Part is constituted by the attempt at going further than the

bipolarity symmetrical-asymmetrical and at considering so-called symmetrical being as a mode of being in which reality 'is lived' as a homogeneous indivisible totality, and in which 'viewing-feeling' and being are one and the same thing. Asymmetrical being, for its part, conceives reality as divisible or formed by parts and, as such, related to spatio-temporality. It seems clear that the psycho-analytical conception of man can follow from the consequences of the interaction between these two modes of being, as formulated in the terms just mentioned.

It is too early to say how successfully such a formulation will resist criticism and further reflection, as well as the confrontation with clinical reality. But at first sight it appears thought-provoking and stimulating in our understanding of clinical reality as revealed by psycho-analysis, and, in fact, the more one reflects about it, the deeper this understanding and the greater its possibilities appear to be.

Part VIII. In this Part an attempt is made to go above the detailed consideration of each subject in order to discover some general lines that seem to emerge from this study as a whole. The purpose of this Part is to complement the detailed view with a more comprehensive view, so that the reader may get a perspective of both the wood and the trees. The problems raised in this Part refer to fundamental questions and some of the assertions made may appear ambitious. The reader will submit them to his critical judgment and if, in the end, he feels that my claims were unjustified, I hope that, at least, they will have furnished him with the occasion for a personal re-examination of these questions and, in this way, also given him the opportunity for contributing to their elucidation.

Chapter 30 must be considered as not belonging to the text of the book proper. It represents *one* personal reaction, out of many, and at a particular time, to the metabolic exchange of the ideas put forward here. After much reflection I have decided to leave it as it is, for, in my opinion, it fulfils a definite, legitimate function in connection with the subjects discussed here.

Part IX. When I revised and corrected the manuscript and made all the additions and clarifications necessary to make my meaning more precise and cohesive, I realised that the subject of multidimensional space was present throughout the book in a far greater degree than I had at first thought. In fact I had deliberately decided not to deal with it in full, hoping to leave it for a future occasion. What has actually happened is that the natural development of the ideas put forward here has brought me, gradually and imperceptibly, to another turn of the spiral which I have followed in my research. This turn corresponds to the studies which I had made long before I came to formulate the principle of symmetry; they deal with the